Regional Inflation Expectations and the New Keynesian Phillips Curve

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Abstract

This paper estimates the New Keynesian Phillips curve (NKPC) using novel regional measures of inflation expectations, addressing the empirical challenge of separately identifying the coefficients on expected inflation and the unemployment gap in the presence of limited aggregate variation. Existing cross-sectional approaches identify a small slope by assuming that long-run expectations are spatially uniform—an assumption ruled out by the literature documenting regional heterogeneity in expectations. We construct synthetic expectation measures from Michigan Survey of Consumers microdata combined with local demographic and industry information across 21 U.S. MSAs, and estimate the regional Phillips curve in pre-COVID and post-COVID samples. Within each sample we compare two-way fixed-effects OLS with IV estimates that instrument unemployment with a Bartik shift-share and the relative intermediate-input price with its 24-month lag. OLS misleads in opposite directions across the two regimes. Pre-COVID, OLS produces a steep slope while IV recovers the flat slope of Hazell et al. (2022) with the expectations channel still positive and significant. Post-COVID, OLS produces a flat slope while IV produces a very steep slope and an insignificant expectations coefficient. The intermediate-input coefficient flips sign between OLS and IV in both regimes. Together, these patterns indicate a regime-dependent regional Phillips curve: flat under credible identification in the pre-COVID period and markedly steeper during the post-COVID inflation surge, with the role of regional inflation expectations itself shifting between regimes.

Keywords: Phillips curve; inflation expectations; regional inflation; small-area estimation; shift-share instrument.

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1 Introduction

The most influential recent estimate of the regional New Keynesian Phillips curve identifies a small slope by assuming that long-run inflation expectations are common across U.S. regions. Hazell et al. (2022) identify the slope of the Phillips curve from regional variation in labor market slack and non-tradable inflation, leaving aggregate shocks and common long-run inflation expectations to be absorbed by the time fixed effects. Their result implies a flat aggregate Phillips curve, with the early-1980s disinflation driven primarily by shifting long-run expectations under Volcker rather than by a steep unemployment-inflation tradeoff. The post-COVID inflation surge of 2021–2023 has put the small slope result under fresh empirical pressure and renewed scrutiny of its identifying assumptions.

The validity of the fixed effects strategy rests on the assumption that inflation expectations do not vary systematically across regions. A parallel literature in household finance and behavioral macroeconomics has established precisely the opposite. Malmendier and Nagel (2016) show that lifetime inflation experiences shape individual expectations through experience-based learning, generating persistent heterogeneity that tracks regional economic histories. Hajdini et al. (2024) document substantial and stable differences in inflation expectations across U.S. metropolitan areas. D'Acunto et al. (2021) provide further evidence that local price environments and demographic composition produce systematic regional divergence in expectations. Together, these findings imply that the expectation component of inflation is far from common across regions, and that the fixed effects strategy may be absorbing something other than what the underlying theory predicts.

The econometric consequences are direct. If regional deviations in inflation expectations are correlated with regional labor market conditions—a plausible outcome under experience-based learning, since metropolitan areas with persistently tight labor markets accumulate higher inflation expectations through repeated exposure to low unemployment and rising wages—then omitting regional expectations from the fixed effects regression generates omitted variable bias in the slope estimate. The direction and magnitude of this bias depend on the joint covariance of expectations and slack with realized inflation. Whether this bias is quantitatively large enough to matter for the central conclusion of Hazell et al. (2022) is an empirical question, and one that bears on the design of monetary policy.

We embed this empirical question in the canonical New Keynesian Phillips curve, $\pi_t = \beta E_t \pi_{t+1} + \kappa \widetilde{mc}_t$ (Galí and Gertler, 1999). Cerrato and Gitti (2022) adapt this framework to a regional setting in which firms face Calvo price-setting frictions and use both labor and intermediate inputs in production. Iterating the structural equation forward yields a

regression specification in which inflation in metropolitan area m at time t depends on local inflation expectations, local unemployment, the relative price of intermediate inputs, and a pair of fixed effects:

$$\pi_{m,t} = \eta_m + \xi_t + \psi\, E_{m,t}[\pi_{t+1}] - \kappa\, u_{m,t} + \delta\, p^x_{m,t} + \varepsilon_{m,t}, \tag{1}$$

with κ the slope of the regional Phillips curve. We test whether incorporating $E_{m,t}[\pi_{t+1}]$ moves the estimate of κ relative to specifications that omit a regional expectation control.

This paper provides the first systematic attempt to address the empirical question using measures of regional inflation expectations. We use four datasets that span both sides of the expectation channel: two on household inflation expectations and two on firm inflation expectations. The Michigan Survey of Consumers (MSC) provides households' point forecasts back to 1978. We exploit FIPS-coded county identifiers in the restricted microdata to assign respondents directly to metropolitan areas. The Federal Reserve Bank of Cleveland's Indirect Consumer Inflation Expectations (ICIE) survey, regionally representative by design, provides a direct post-2021 household measure that does not require small-area estimation. The Cleveland Fed's Survey of Firms' Inflation Expectations (SoFIE) supplies firm CPI forecasts from a nationally representative panel of CEOs from 2018 onward. The Atlanta Fed's Business Inflation Expectations (BIE) survey supplies firms' unit-cost expectations as a complementary measure of price-setting margins. Demographic cell counts come from the American Community Survey, regional industry shares from the Quarterly Census of Employment and Wages (QCEW), and local consumer price index (CPI) and unemployment from the Bureau of Labor Statistics (BLS) at the MSA level.

Our identification strategy is a Bayesian hierarchical first stage paired with a rolling-decennial shift-share instrumental-variables second stage. The first stage uses two *separate* multilevel models so that the instrument and the endogenous regressor are statistically independent objects. A national-only model estimates demographic-cell time predictions $\hat{\pi}^e_{c,t}$ from individual MSC microdata with no MSA-level effects, where cells are defined by interactions of age, education, income quartile, gender, and race; these national predictions enter only into the instrument. A separate MSA-level multilevel model adds MSA random effects, time effects, and an MSA-by-time random effect $w_{m,t}$ that follows an AR(1) process in time and a CAR prior across geographically adjacent metropolitan areas (Marhuenda et al., 2013), post-stratified to ACS MSA cell counts to produce the regressor $\hat{\theta}_{m,t}$ (Park et al., 2004). The MSA-by-time component $w_{m,t}$ is the methodological centerpiece: it is a modeled object rather than a residual that gets absorbed by the second-stage fixed effects, and the spatio-temporal priors borrow strength across adjacent quarters and neighboring

MSAs to deliver well-identified MSA-specific dynamics despite thin per-MSA-quarter survey samples. The second stage uses a Bartik-style shift-share instrument constructed as $Z^{\pi}_{m,t} = \sum_c \omega^{d(t)}_{c,m} \hat{\pi}^{e}_{c,t}$, where $\omega^{d(t)}_{c,m}$ is MSA m's cell share from the most recent prior decennial Census ($d(t) \in \{1980, 1990, 2000, 2010, 2020\}$). The exclusion restriction requires decennial shares to be uncorrelated with MSA-level Phillips curve residuals after fixed effects, the unemployment instrument, and the intermediate-input control; we engage with this assumption following the share-by-share framework of Goldsmith-Pinkham et al. (2020) and a falsification pre-test on observable residuals.

As preliminary evidence pending the implementation of this full strategy, we report results from a transparent first-pass synthetic projection of MSC microdata onto MSA-level demographic distributions. We construct three first-pass variants that differ in the information set used in the first stage: a baseline measure built from demographic composition alone, a house-prices-only variant that drops demographics altogether, and an augmented measure that supplements the baseline with MSA-level house price inflation. We use all three to estimate equation (1) on a monthly panel of 21 MSAs, splitting the sample into a pre-COVID period (January 2004–February 2020) and the post-COVID inflation surge (March 2021–June 2022). Within each period we compare two-way fixed-effects OLS estimates with an instrumental-variables specification that instruments unemployment with a Bartik-style sectoral labor demand shock $Z^{u}_{m,t}$ and the relative intermediate-input price $p^{x}_{m,t}$ with its 24-month lagged value, leaving the regional expectation regressor as an OLS variable.

The headline finding is that OLS misleads in opposite directions across the two regimes. In the pre-COVID sample, OLS on the baseline demographic measure produces a steep regional Phillips curve ($\hat{\kappa} = -0.225$, significant at the 1% level) and a positive, significant role for expectations ($\hat{\psi} = 0.272$, $p < 0.01$); the IV estimates recover a flat slope ($\hat{\kappa} = 0.098$, statistically indistinguishable from zero) consistent with the central finding of Hazell et al. (2022), while the expectations channel remains positive and significant ($\hat{\psi} = 0.154$, $p < 0.01$). In the post-COVID sample, the pattern reverses: OLS produces a flat slope ($\hat{\kappa} = -0.028$, insignificant) and a strong expectations channel ($\hat{\psi} = 0.447$, $p < 0.01$); the IV estimates produce a very steep slope ($\hat{\kappa} = -1.338$, $p < 0.01$) and an expectations coefficient indistinguishable from zero ($\hat{\psi} = 0.104$). The same qualitative pattern holds across the housing-augmented and demographics-augmented first-pass variants. The intermediate-input price coefficient $\hat{\delta}$ changes sign between OLS and IV in both regimes, from negative under OLS to positive under IV; the positive IV sign aligns with the Cerrato–Gitti structural restriction that δ is the intermediate-input share, while the negative OLS sign suggests that the

cost-push channel is contaminated by reverse causation from contemporaneous regional inflation shocks that simultaneously raise local prices and depress upstream input demand. The central message of the preliminary results is the quantitative importance of instrumenting both unemployment and the intermediate-input price; the natural next step, which the full Bayesian hierarchical IV implementation is designed to deliver, is to instrument regional inflation expectations as well.

This paper is most directly related to two strands of the literature. The first is the empirical literature on regional Phillips curves (Babb and Detmeister, 2017; Beraja et al., 2019; McLeay and Tenreyro, 2020; Hooper et al., 2020; Hazell et al., 2022; Cerrato and Gitti, 2022; Fitzgerald et al., 2024; Gitti, 2024). We are the first to estimate the regional Phillips curve while directly controlling for regional inflation expectations rather than absorbing their variation into time fixed effects, and we are the first to instrument the expectation channel using a regional shift-share constructed from demographic composition. The second is the literature on the regional structure of inflation expectations (Malmendier and Nagel, 2016; Hajdini et al., 2024; D'Acunto et al., 2021), to which we contribute a synthetic estimation framework that produces high-frequency, geographically granular measures of expectations from a survey designed to be representative only at the national level.

The remainder of the paper is organized as follows. Section 2 sets out the structural Phillips curve framework. Section 3 describes the data. Section 4 develops the empirical strategy: the regression specification, the full Bayesian hierarchical methodology and shift-share IV, and the preliminary first-pass implementation. Section 5 reports preliminary OLS and IV estimates of the regional Phillips curve across the three first-pass expectation measures and benchmarks them against Hazell et al. (2022). Section 6 concludes.

2 Theoretical Framework

We anchor the empirical specification in the canonical New Keynesian Phillips curve, adapted to a regional setting in which firms use labor and intermediate inputs in production. The structural parent is the NKPC of Galí and Gertler (1999), in which monopolistically competitive firms set prices subject to Calvo frictions. Aggregate inflation evolves as

$$\pi_t = \beta\, E_t \pi_{t+1} + \kappa\, \widetilde{mc}_t, \tag{2}$$

where $\beta \in (0, 1)$ is the household discount factor, $\widetilde{mc}_t$ is the log deviation of marginal cost from steady state, and κ is a function of the elasticity of substitution across goods and the

Calvo price-reset probability. At quarterly frequency the discount factor is conventionally calibrated to $\beta \approx 0.99$.

Cerrato and Gitti (2022) adapt equation (2) to a regional setting in which firms in metropolitan area m produce output using a Cobb–Douglas technology over labor and intermediate inputs. Marginal cost in the regional setting depends on local labor market conditions through wages and on the price of intermediate inputs purchased from upstream sectors. Iterating equation (2) forward and substituting the regional marginal cost expression delivers the regression equation

$$\pi_{m,t} = \eta_m + \xi_t + \psi\, E_{m,t}[\pi_{t+1}] - \kappa\, u_{m,t} + \delta\, p^x_{m,t} + \varepsilon_{m,t}, \tag{3}$$

where $\pi_{m,t}$ is local CPI inflation, $E_{m,t}[\pi_{t+1}]$ is short-run expected inflation in MSA m, $u_{m,t}$ is the local unemployment rate, $p^x_{m,t}$ is the relative price of intermediate inputs faced by local firms, and (η_m, ξ_t) are MSA and time fixed effects.

The fixed effects play distinct roles in identification. The MSA fixed effect η_m absorbs persistent differences across metropolitan areas in the level of the local Phillips curve relationship—differences in industrial structure, local wage-setting institutions, time-invariant components of consumption baskets. The time fixed effect ξ_t absorbs aggregate shocks common to all metropolitan areas, including national monetary and fiscal policy responses. Hazell et al. (2022) show, by iterating the structural Phillips curve forward to express the equation in terms of the entire path of expected future marginal costs, that ξ_t also absorbs common long-run inflation expectations anchored to the prevailing monetary policy regime. By directly including a measure of short-run expectations $E_{m,t}[\pi_{t+1}]$ alongside the time fixed effects, we isolate the role of *regional heterogeneity* in expectations over and above the common aggregate component that ξ_t absorbs.

The structural mapping pins down what we should expect for each parameter. The coefficient on expectations satisfies $\psi = \beta$, so under standard calibrations ψ is close to but slightly below one; departures of $\hat{\psi}$ from β are jointly informative about violations of the Calvo and constant-elasticity-of-substitution assumptions of the structural model and about the role of the expectations channel, and are not separately identified without additional structure. The coefficient κ is the slope of the regional Phillips curve, an object of direct interest for monetary policy. The coefficient δ is the intermediate-input share in production, which under the Cobb–Douglas assumption equals the Cobb–Douglas exponent on intermediate inputs and is therefore positive at the structural target. The relative price of intermediate inputs $p^x_{m,t}$ is constructed from BLS Producer Price Indices weighted by MSA-level industry shares.

Including $p^x_{m,t}$ in the regression is essential: it absorbs cost-push variation in inflation that would otherwise correlate with both unemployment (through factor demand) and expectations (through cost pass-through), and that omission would contaminate the identification of κ and ψ.

3 Data

The empirical analysis combines three classes of data: regional measures of household and firm inflation expectations from four micro-level surveys; demographic and industry composition data from the American Community Survey and the Quarterly Census of Employment and Wages; and regional CPI and unemployment data from the Bureau of Labor Statistics. This section describes each source and summarizes the resulting analysis sample.

3.1 Michigan Survey of Consumers

The Michigan Survey of Consumers (MSC) is the longest-running survey of household inflation expectations in the United States, conducted monthly by the University of Michigan Survey Research Center since 1978. Each wave interviews approximately 500 households drawn from a national probability sample. The survey elicits a numeric one-year-ahead inflation expectation through the question: "By about what percent do you expect prices to go up/down on the average, during the next 12 months?" Respondents report a number between -95 and $+95$, which we use as the dependent variable in our first-stage regressions. The survey also asks a parallel five-year-ahead question, which we reserve for an analysis of long-run expectations.

The MSC's primary advantage for this paper is the availability of both individual-level demographic characteristics and geographic identifiers in the restricted microdata. Demographic variables include age, education, income quartile, gender, race, marital status, and homeownership—all of which are strong predictors of the level of reported expectations (D'Acunto et al., 2021). Geographic identifiers include FIPS-coded county and metropolitan statistical area codes, which we use to assign respondents to states and MSAs. Access to these identifiers is governed by a data-use agreement with the Survey Research Center.

The principal limitation of the MSC for this paper is sample size at the regional level. With approximately 500 respondents per month split across 50 states and several hundred MSAs, monthly per-MSA cells contain on average a handful of observations and many cells

are empty. Direct within-MSA aggregation of MSC responses is therefore both noisy and unrepresentative of the underlying regional population, motivating the small-area estimation strategy described in Section 4. Table 1 reports summary statistics for the demographic variables in the MSC.

Table 1: Summary statistics: Michigan Survey of Consumers

Variable	MSC
Age	49 years
Female	54%
Low income	43%
Middle income	33%
High income	25%
< High school	4%
High school	6%
Some college	28%
N	277,160

Notes. Summary statistics for demographic variables in the Michigan Survey of Consumers, 1978–2024. Survey weights applied.

3.2 Other Survey Data on Inflation Expectations

We supplement the MSC with three additional surveys that fill gaps in the MSC's coverage of either regional representativeness or firm expectations.

The Federal Reserve Bank of Cleveland's *Indirect Consumer Inflation Expectations* (ICIE) survey, launched in 2021, is the only U.S. household survey designed to be representative at sub-national geographies. The survey elicits inflation expectations indirectly through questions about household income needs, an approach that mitigates the framing effects associated with direct numeric elicitation (Hajdini et al., 2024). ICIE provides a direct post-2021 household measure of regional inflation expectations that does not require small-area estimation, and we use it both as a non-shift-share check on our full Bayesian hierarchical estimates and as a validation target for the synthetic measures constructed from MSC data.

The Federal Reserve Bank of Cleveland's *Survey of Firms' Inflation Expectations* (SoFIE) is a quarterly nationally representative panel of CEOs that has elicited firm CPI forecasts and

subjective probability distributions over inflation since 2018. The SoFIE provides the firm-side analogue to the MSC household data and is the primary firm CPI-expectations measure in the analysis. The Federal Reserve Bank of Atlanta's *Business Inflation Expectations* (BIE) survey, in operation since 2011, provides high-frequency sector-level expectations of own-firm unit cost growth from approximately 640 firms in the Sixth Federal Reserve District. Because BIE elicits unit cost growth rather than CPI inflation, it is not directly comparable to the structural object $E_{m,t}[\pi_{t+1}]$ in equation (3); we use it in a complementary analysis of price-setting margins rather than as a direct measure of firm CPI expectations.

3.3 Demographic and Industry Composition

Demographic cell counts at the MSA level come from the American Community Survey (ACS), accessed through IPUMS. The ACS provides annual measures of the joint distribution of age, education, income, gender, race, marital status, and homeownership for each MSA. We construct demographic cells defined by deep interactions of these characteristics, with the interaction structure following Ghitza and Gelman (2013), and use the resulting cell counts $N_{c,m}$ as post-stratification weights in the second stage of the multilevel-regression-and-post-stratification (MRP) procedure described in Section 4. We additionally use rolling decennial Census shares (from 1980, 1990, 2000, 2010, and 2020) as the share weights in our shift-share instrument for inflation expectations, with each observation in our panel using the cell shares from the most recent prior decennial Census.

Regional industry composition comes from the BLS Quarterly Census of Employment and Wages (QCEW), which reports MSA-level employment and establishment counts at the NAICS-2-digit level on a quarterly basis. We use 2010 NAICS-2-digit MSA employment shares to construct the relative price of intermediate inputs $p^x_{m,t}$ from sectoral Producer Price Indices, following Cerrato and Gitti (2022), and to build the unemployment instrument from sectoral labor demand shocks.

3.4 Regional CPI and Unemployment

Regional CPI data come from the BLS, which publishes monthly or bi-monthly headline CPI for 21 metropolitan statistical areas beginning in January 1988. We compute 12-month CPI inflation $\pi_{m,t}$ for each MSA from these series; this is the dependent variable in the Phillips curve regression. MSA-level unemployment rates are from the BLS Local Area Unemployment Statistics (LAUS) program, available beginning in January 1990. The intersection of

these two series defines the empirical sample.

Our analysis sample is split into two regimes that reflect the structural break in U.S. inflation dynamics around the pandemic. The pre-COVID sample covers January 2004 to February 2020 across the 21 MSAs for which BLS direct CPI is available, yielding a balanced panel of approximately 2,500 MSA-month observations in the baseline specification. The post-COVID inflation surge sample covers March 2021 to June 2022 and contains 320 MSA-month observations. Specifications that include the housing-augmented expectation measure use the FHFA all-transactions house price index, available quarterly for all 21 MSAs in our sample beginning in 2007 and interpolated linearly to monthly frequency, and yield slightly smaller pre-COVID samples (approximately 2,080–2,160 observations).

4 Empirical Strategy

This section sets out the empirical strategy in four parts. Section 4.1 states the Phillips curve regression. Section 4.2 describes the full Bayesian hierarchical methodology that the paper plans to implement: a two-model first stage that produces both an instrument and an endogenous regressor as statistically independent objects, with the regressor enriched by an MSA-by-time random effect under spatio-temporal smoothing. Section 4.3 describes the shift-share instrumental-variables strategy that pairs with the Bayesian first stage. Section 4.4 describes a transparent first-pass implementation, in three variants, whose preliminary OLS and IV estimates we report in Section 5.

4.1 Phillips Curve Specification

We estimate the regional New Keynesian Phillips curve in equation (3), reproduced here for convenience:

$$\pi_{m,t} = \eta_m + \xi_t + \psi\, E_{m,t}[\pi_{t+1}] - \kappa\, u_{m,t} + \delta\, p^x_{m,t} + \varepsilon_{m,t}. \tag{4}$$

The dependent variable $\pi_{m,t}$ is 12-month CPI inflation in MSA m at month t. The MSA fixed effects η_m absorb time-invariant local characteristics; the time fixed effects ξ_t absorb aggregate shocks common to all metropolitan areas, including, by the iteration argument of Hazell et al. (2022), common long-run inflation expectations anchored to the prevailing monetary policy regime. The structural Cerrato–Gitti restrictions imply $\psi = \beta \approx 0.99$, κ is the slope of the Phillips curve, and δ is the (positive) intermediate-input share. Estimation by OLS recovers $\hat{\psi}$, $\hat{\kappa}$, and $\hat{\delta}$ that are consistent for the structural parameters only if (i) the

regressors are measured without error, (ii) they are uncorrelated with the residual $\varepsilon_{m,t}$, and (iii) reverse causation from contemporaneous regional inflation shocks to unemployment and intermediate-input prices is absent. Each of these conditions is restrictive, and our empirical strategy is designed to relax them through the IV second stage of Section 4.3.

4.2 Constructing Regional Inflation Expectations

Direct within-MSA aggregation of MSC responses fails as a measurement strategy: the regional sub-samples of the MSC are too small to support stable aggregation, and the MSC's national-level sampling design does not deliver representativeness within metropolitan areas. We therefore adopt a two-model Bayesian hierarchical approach that combines small-area estimation with multilevel-regression-and-post-stratification (MRP). The two-model structure is essential to the identification strategy of Section 4.3: it ensures that the instrument and the endogenous regressor in the second-stage regression are statistically independent objects rather than joint summaries of a single posterior.

The first model is a *national-only* multilevel model that estimates demographic-cell time predictions $\hat{\pi}^e_{c,t}$ from individual MSC microdata, with no MSA-level effects. Cells c are defined by interactions of age, education, income quartile, gender, and race—approximately 640 cells in our preferred specification, with marital status and homeownership entering as additive covariates. The model takes the form

$$\pi^e_{i,t} = \alpha_t + \mu_{c[i],t} + X_i'\zeta + \varepsilon_{i,t}, \tag{5}$$

where $\pi^e_{i,t}$ is individual i's reported one-year-ahead inflation expectation in month t, $\mu_{c,t}$ is a time-varying demographic-cell random intercept, and X_i collects the additive covariates. The cell predictions $\hat{\pi}^e_{c,t}$ are the posterior means of $\alpha_t + \mu_{c,t}$. These predictions feed only into the construction of the shift-share instrument described in Section 4.3; they do not enter the regressor.

The second model is an *MSA-level* multilevel model that produces the regressor $\hat{\theta}_{m,t}$ used in equation (4). The model has the same demographic-cell structure as the national-only model but adds MSA random effects δ_m, time effects τ_t, and an MSA-by-time random effect $w_{m,t}$ that follows an AR(1) process in time and a conditional autoregressive (CAR) prior

across geographically adjacent MSAs:

$$\pi^e_{i,m,t} = \alpha + \mu_{c[i]} + \delta_m + \tau_t + w_{m,t} + X_i'\zeta + \varepsilon_{i,m,t}, \tag{6}$$

$$w_{m,t} = \rho\, w_{m,t-1} + \nu_{m,t}, \qquad \nu_{\cdot,t} \sim \mathrm{CAR}(\lambda, W), \tag{7}$$

where W is the geographic adjacency matrix among the 21 MSAs in our sample and λ controls the strength of spatial smoothing. The MSA-by-time component $w_{m,t}$ is the methodological centerpiece: it is a modeled object rather than a residual that gets absorbed by the second-stage fixed effects in equation (4), and the spatio-temporal priors borrow strength across adjacent quarters and neighboring MSAs to deliver well-identified MSA-specific dynamics despite thin per-MSA-quarter survey samples (Marhuenda et al., 2013).

The regressor $\hat{\theta}_{m,t}$ is obtained by post-stratifying the cell-level posterior predictions to ACS MSA cell counts (Park et al., 2004):

$$\hat{\theta}_{m,t} = \frac{\sum_c N_{c,m}\, \hat{\pi}^e_{c,m,t}}{\sum_c N_{c,m}}, \tag{8}$$

where $N_{c,m}$ is the ACS count of MSA-m residents in demographic cell c and $\hat{\pi}^e_{c,m,t}$ is the posterior mean of $\alpha + \mu_c + \delta_m + \tau_t + w_{m,t} + \bar{X}_c'\zeta$. The post-stratification step adjusts for differences in the demographic composition of MSC respondents across MSAs (using ACS as the population ground truth) and shrinks thin MSA-time cells toward the demographic-conditional mean implied by the multilevel model.

Both models are estimated by Hamiltonian Monte Carlo (HMC) with 4,000 posterior draws, used to propagate measurement uncertainty into the second-stage Phillips curve regression. Variational inference serves as a computationally cheaper fallback if HMC does not converge in feasible time. As model diagnostics, we report the posterior versus prior standard deviation of the MSA-by-time random effect as a measure of the data's contribution to the spatio-temporal component, and we compare the post-stratified MSA expectation estimates against direct sample means in the largest MSAs (New York, Los Angeles, Chicago) where direct estimation is feasible.

4.3 Identification: Shift-Share Instrumental Variables

Even with a credible measurement of regional expectations, the OLS estimate of equation (4) confounds the causal effects of expectations, unemployment, and intermediate-input prices on inflation with reverse causation from contemporaneous regional inflation shocks. The pre-

liminary results of Section 5 show that this contamination is quantitatively important: under OLS the unemployment slope and the intermediate-input coefficient are biased in opposite directions across the pre- and post-COVID regimes, and the bias on the intermediate-input coefficient is large enough that its sign flips between OLS and IV. We address these biases with a shift-share instrumental-variables strategy paired with a 24-month-lagged instrument for the intermediate-input price.

Instrument for unemployment. A consistent estimate of the slope κ requires accounting for regional supply shocks that are correlated with local unemployment but enter the residual through channels other than expectations and intermediate-input prices. Following Cerrato and Gitti (2022), we instrument $u_{m,t}$ with a Bartik-style sectoral labor demand shock that interacts *beginning-of-period* MSA NAICS-2-digit industry shares with national sectoral employment growth in the intermediate-input sectors:

$$Z^{u}_{m,t} = \sum_{j} e^{0}_{j,m} \, \Delta L^{\text{nat}}_{j,t}, \tag{9}$$

where $e^{0}_{j,m}$ is MSA m's NAICS-2-digit employment share in sector j at the beginning of the analysis sample (2004) and $\Delta L^{\text{nat}}_{j,t}$ is national sectoral employment growth excluding MSA m.

Instrument for the intermediate-input price. We instrument $p^{x}_{m,t}$ with its 24-month lagged value, $p^{x}_{m,t-24}$. The lag length is chosen to ensure that the instrument is uncorrelated with contemporaneous regional inflation shocks while remaining a strong predictor of current input prices through the persistent component of MSA-level industry composition.

Instrument for expectations. The instrument for $E_{m,t}[\pi_{t+1}]$ is constructed from MSA m's rolling decennial Census cell shares interacted with the national demographic-cell time predictions $\hat{\pi}^{e}_{c,t}$ from the national-only first-stage model:

$$Z^{\pi}_{m,t} = \sum_{c} \omega^{d(t)}_{c,m} \, \hat{\pi}^{e}_{c,t}, \tag{10}$$

where $\omega^{d(t)}_{c,m}$ is MSA m's population share in demographic cell c from the most recent prior decennial Census, $d(t) \in \{1980, 1990, 2000, 2010, 2020\}$. Rolling decennial shares avoid the forward-looking-share problem that arises when fixed-period shares are applied to observations earlier than the share-construction date. The Bartik exclusion restriction requires

that the decennial cell shares are uncorrelated with the regional Phillips curve residual after fixed effects, the unemployment instrument, and the intermediate-input control (Borusyak et al., 2022). We engage with this assumption following the share-by-share framework of Goldsmith-Pinkham et al. (2020): we report which demographic shares carry the bulk of the identifying weight in the IV first stage and pre-test the assumption by regressing decennial shares on observable Phillips curve residuals as a falsification check. The preliminary results of Section 5 use $Z^u_{m,t}$ and $p^x_{m,t-24}$ as instruments for unemployment and the intermediate-input price respectively, while leaving the regional expectation regressor as an OLS variable; instrumenting expectations with $Z^\pi_{m,t}$ requires the full Bayesian hierarchical first stage of Section 4.2 and is the central next step we leave for the implementation of the full strategy.

Bayesian IV regression. We integrate the first-stage measurement uncertainty in $\hat{\theta}_{m,t}$ with the second-stage 2SLS uncertainty by estimating equation (4) as a Bayesian IV regression, in which the joint posterior on (ψ, κ, δ) is obtained by drawing the regressor $\hat{\theta}_{m,t}$ from its first-stage posterior and re-estimating the second stage at each draw. We report partial F-statistics on each instrument as a check on first-stage relevance.

4.4 Preliminary First-Pass Implementation

Because the full Bayesian hierarchical first stage and Bayesian IV regression are computationally substantial, we report preliminary OLS and IV evidence using a transparent first-pass synthetic projection in place of the Bayesian-MRP regressor. The first-pass measure is constructed in three variants that decompose the contributions of demographic composition and local housing dynamics to regional expectations: a demographic-only baseline that captures persistent cross-MSA heterogeneity, a housing-only variant that captures high-frequency local price pressures, and a combined specification that incorporates both. We use all three variants to estimate equation (4) in Section 5 and treat their consistency—or divergence—as informative about the sources of identifying variation.

4.4.1 Baseline: demographic composition

The baseline first-pass measure adopts a two-step OLS procedure. In the first stage, we exploit the MSC's national representativeness to estimate a cross-sectional model of individual inflation expectations as a function of observable demographic characteristics, separately in

each month:

$$E_{i,t}[\pi_{t+1}] = \beta_{0t} + \beta_{1t}\,\text{Education}_i + \beta_{2t}\,\text{Gender}_i + \beta_{3t}\,\text{MaritalStatus}_i + \beta_{4t}\,\text{Age}_i + \varepsilon_{i,t}, \quad (11)$$

where Education, Gender, and Marital Status are categorical and Age is continuous. The monthly indexing of the coefficients allows the demographic gradient of expectations to vary at the national level over time.

In the second stage, we project the estimated coefficients onto MSA-level demographic distributions from the ACS to generate an MSA-level expectation series:

$$\hat{E}_{m,t}[\pi_{t+1}] = \hat{\beta}_{0t} + \hat{\beta}_{1t}\,\overline{\text{Education}}_{m,t} + \hat{\beta}_{2t}\,\overline{\text{Gender}}_{m,t} + \hat{\beta}_{3t}\,\overline{\text{MaritalStatus}}_{m,t} + \hat{\beta}_{4t}\,\overline{\text{Age}}_{m,t}, \quad (12)$$

where the overline denotes the ACS MSA-level mean. Because the ACS demographic averages vary annually while the first-stage coefficients vary monthly, the resulting first-pass series captures both time-varying shifts in the national demographic gradient and persistent cross-sectional differences in MSA demographic composition.

The baseline first-pass measure has the merit of transparency—it can be reproduced from public data with minimal modeling assumptions—but it has three limitations that the full Bayesian hierarchical methodology of Section 4.2 addresses. First, the cross-sectional variation is driven entirely by time-invariant or slow-moving MSA demographic means, with no MSA-specific dynamics analogous to the $w_{m,t}$ random effect of the full methodology. Second, the demographic-projection structure assumes that the national demographic gradient of expectations applies within each MSA—a homogeneity-of-coefficients assumption that the cited literature on experience-based learning questions. Third, the OLS second-stage regression of inflation on the first-pass measure does not address the endogeneity of expectations, which the IV strategy of Section 4.3 is designed to resolve.

4.4.2 Using House Prices to Capture Regional Effects

The house-prices-only variant constructs the first-pass measure using only local housing dynamics in the first stage, dropping demographics altogether. This isolates the contribution of housing to regional inflation expectations and provides a high-frequency, locally-driven alternative to the demographic baseline.

In the first stage, we regress individual MSC inflation expectations on individual house

price expectations alone:

$$E_{i,t}[\pi_{t+1}] = \beta_0 + \beta_1 \, E_{i,t}[\pi^h_{t+1}] + \varepsilon_{i,t}, \tag{13}$$

where $E_{i,t}[\pi^h_{t+1}]$ is individual i's one-year-ahead house price expectation, elicited directly in the MSC.

In the second stage, we replace $E_{i,t}[\pi^h_{t+1}]$ with realized MSA-level house price inflation $\pi^h_{m,t}$ from the FHFA all-transactions house price index:

$$\hat{E}_{m,t}[\pi_{t+1}] = \hat{\beta}_0 + \hat{\beta}_1 \, \pi^h_{m,t}. \tag{14}$$

The substitution of realized for expected house price inflation is motivated by the high serial correlation of local housing markets: MSA-level house price growth is highly persistent, so realized inflation is a strong predictor of one-year-ahead expectations. The FHFA index is available quarterly for all 21 MSAs in our sample beginning in 2007; we interpolate linearly to monthly frequency. The housing-only variant introduces a source of variation that is both local and high-frequency, unlike the demographic baseline, which varies only with slow-moving MSA demographic means.

4.4.3 Combining Demographics and Regional Information

The combined first-pass variant enriches the demographic baseline by adding house price inflation to the first-stage regression, capturing both persistent cross-sectional heterogeneity from demographics and high-frequency local variation from housing.

In the first stage, we estimate

$$E_{i,t}[\pi_{t+1}] = \beta_0 + \beta_1 \, \text{Education}_i + \beta_2 \, \text{Gender}_i + \beta_3 \, \text{MaritalStatus}_i + \beta_4 \, \text{Age}_i + \beta_5 \, E_{i,t}[\pi^h_{t+1}] + \varepsilon_{i,t}, \tag{15}$$

where Education, Gender, and Marital Status are categorical, Age is continuous, and $E_{i,t}[\pi^h_{t+1}]$ is the individual house price expectation from the MSC.

In the second stage, we project the estimated coefficients onto MSA-level demographic means from the ACS and replace $E_{i,t}[\pi^h_{t+1}]$ with realized MSA-level house price inflation $\pi^h_{m,t}$:

$$\hat{E}_{m,t}[\pi_{t+1}] = \hat{\beta}_0 + \hat{\beta}_1 \, \overline{\text{Education}}_{m,t} + \hat{\beta}_2 \, \overline{\text{Gender}}_{m,t} + \hat{\beta}_3 \, \overline{\text{MaritalStatus}}_{m,t} + \hat{\beta}_4 \, \overline{\text{Age}}_{m,t} + \hat{\beta}_5 \, \pi^h_{m,t}, \tag{16}$$

where the overline denotes the ACS MSA-level mean. The combined first-pass measure

incorporates both persistent demographic differences across MSAs and local housing market dynamics, providing a richer characterization of regional expectations than either input alone.

5 Preliminary Phillips Curve Estimates

This section reports preliminary estimates of the regional Phillips curve in equation (4), using each of the first-pass synthetic expectation measures developed in Section 4.4. We split the analysis sample into a pre-COVID period (January 2004–February 2020) and the post-COVID inflation surge (March 2021–June 2022), reflecting the structural break in U.S. inflation dynamics around the pandemic. Within each period we report two specifications: two-way fixed-effects OLS, and an instrumental-variables specification that instruments the unemployment rate with the Bartik-style sectoral labor demand shock $Z^u_{m,t}$ in equation (9) and the relative intermediate-input price with its 24-month lagged value. The regional inflation expectation regressor is not instrumented in the preliminary results; we view instrumenting expectations as the central next step that the full Bayesian hierarchical IV strategy of Section 4.2 is designed to deliver.

The headline finding, which holds across all three first-pass variants, is that the OLS estimates misrepresent the regional Phillips curve in opposite directions across the two regimes. Under OLS, the unemployment slope is too steep in the pre-COVID period and too flat in the post-COVID period; instrumenting unemployment recovers a flat slope in the pre-COVID period—consistent with Hazell et al. (2022)—and a much steeper slope in the post-COVID period. The intermediate-input coefficient also changes sign between OLS and IV in both regimes, with the negative OLS sign indicating reverse causation from contemporaneous regional inflation shocks to upstream input prices. The expectations coefficient remains positive and statistically significant under both estimators in the pre-COVID period; in the post-COVID period it is significant under OLS and indistinguishable from zero under IV. We caution that the change in the expectations coefficient between OLS and IV is not itself a clean causal estimate of the expectations channel, since expectations are not instrumented in the preliminary results; instrumenting expectations is the central next step.

5.1 Baseline results

Table 2 reports estimates of equation (4) using the baseline demographic-only synthetic expectation measure $\hat{E}_{m,t}[\pi_{t+1}]$ from equation (12). Columns (1) and (3) report two-way

fixed-effects OLS estimates for the pre- and post-COVID periods, respectively; columns (2) and (4) report the corresponding IV specifications.

Table 2: Baseline Phillips curve: demographic expectations

	Pre Period (2004–2020)		Post Period (2021m3–2022m6)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
$u_{m,t}$	-0.225***	0.098	-0.028	-1.338***
	(0.029)	(0.124)	(0.081)	(0.265)
$E_{m,t}[\pi_{t+1}]$	0.272***	0.154***	0.447***	0.104
	(0.045)	(0.047)	(0.046)	(0.069)
$p^x_{m,t}$	-0.082***	0.166***	-0.172***	0.144**
	(0.014)	(0.033)	(0.040)	(0.069)
Observations	2,468	2,228	320	320
Time FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
F-stat ($Z^u_{m,t}$)	—	55.19	—	40.34

Notes. OLS (columns 1 and 3) and IV (columns 2 and 4) estimates of equation (4). The dependent variable is the 12-month MSA-level CPI inflation rate $\pi_{m,t}$. Inflation expectations $E_{m,t}[\pi_{t+1}]$ are demographic-only synthetic expectations from equation (12). $u_{m,t}$ is the negative log employment rate. $p^x_{m,t}$ is the log ratio of the manufacturing PPI to the CPI all items. In the IV specification, $u_{m,t}$ is instrumented by the Bartik shift-share $Z^u_{m,t}$ in equation (9); $p^x_{m,t}$ is instrumented by its 24-month lag. The OLS specification absorbs MSA and year-quarter fixed effects. The IV specification absorbs MSA fixed effects and year-quarter × initial industry-share time effects. Standard errors clustered at the MSA × year-quarter level. The pre period covers 2004m1–2020m2; the post period covers 2021m3–2022m6. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The pre-COVID OLS estimates in column (1) appear to deliver a textbook regional Phillips curve: the unemployment coefficient is $\hat{\kappa} = -0.225$ (significant at the 1% level), and the expectations coefficient is $\hat{\psi} = 0.272$ (also significant at the 1% level). The intermediate-input coefficient, however, is $\hat{\delta} = -0.082$—negative and significant, opposite to the positive sign implied by the Cerrato–Gitti structural restriction that δ is the intermediate-input share. The negative OLS estimate of δ is the first signal that OLS is contaminated: it indicates that, conditional on unemployment and expectations, MSAs with higher manufacturing PPI inflation experience *lower* CPI inflation, which is implausible from a structural standpoint and points to reverse causation from contemporaneous regional inflation shocks to upstream input prices.

The pre-COVID IV estimates in column (2) confirm that interpretation. Once unemploy-

ment is instrumented with the Bartik shift-share $Z^u_{m,t}$ and the intermediate-input price is instrumented with its 24-month lag, the unemployment coefficient drops in absolute value to $\hat{\kappa} = 0.098$ and is statistically indistinguishable from zero, recovering the flat slope reported by Hazell et al. (2022) in their preferred state-level specification. The intermediate-input coefficient flips sign and becomes positive and significant ($\hat{\delta} = 0.166$, $p < 0.01$), consistent with the structural prediction. The expectations coefficient remains positive and significant ($\hat{\psi} = 0.154$, $p < 0.01$), indicating that regional household inflation expectations have a quantitatively meaningful association with realized regional inflation that is not driven by the OLS bias on unemployment. The first-stage F-statistic on the unemployment instrument is 55.19, well above conventional weak-instrument thresholds.

The post-COVID estimates in columns (3)–(4) reveal the opposite OLS-IV pattern. OLS produces a flat slope ($\hat{\kappa} = -0.028$, insignificant) and a strong expectations channel ($\hat{\psi} = 0.447$, $p < 0.01$); IV produces a very steep slope ($\hat{\kappa} = -1.338$, $p < 0.01$) and an expectations coefficient indistinguishable from zero ($\hat{\psi} = 0.104$). The intermediate-input coefficient flips sign between estimators in the same direction as in the pre-COVID period, from $\hat{\delta} = -0.172$ under OLS to $\hat{\delta} = 0.144$ under IV. The first-stage F-statistic on the unemployment instrument is 40.34. The post-COVID IV slope is roughly an order of magnitude larger in absolute value than the pre-COVID IV slope, consistent with a steepening of the regional Phillips curve during the inflation surge—a finding that mirrors the regime-switching evidence of Cerrato and Gitti (2022) and Gitti (2024).

The change in the expectations coefficient between OLS and IV in the post-COVID period merits caution. Because $E_{m,t}[\pi_{t+1}]$ is not instrumented in the preliminary results, the change in $\hat{\psi}$ between columns (3) and (4) is not a clean causal estimate of the expectations channel. When the IV strategy recovers a credible coefficient on $u_{m,t}$ and $p^x_{m,t}$ that absorbs more of the inflation variation, the OLS-residual variation that the regression attributes to expectations under OLS is mechanically reduced, even if the true causal effect of expectations on inflation has not changed. Disentangling the true expectations channel from the bias in the unemployment and intermediate-input coefficients requires instrumenting $E_{m,t}[\pi_{t+1}]$ with the Bartik shift-share $Z^\pi_{m,t}$ in equation (10), which the full Bayesian hierarchical implementation of Section 4.2 delivers.

5.2 Local-area results

We now turn to the local-area expectation measures, which incorporate MSA-level housing market conditions into the synthetic index. House prices are a natural candidate for enrich-

ing the first-stage projection: shelter is the single largest component of the CPI, housing markets differ substantially and persistently across metropolitan areas, and a growing literature documents that local house price dynamics shape household inflation perceptions (Hajdini et al., 2024). We consider two specifications, the first with house prices alone in the first stage and the second with house prices alongside demographic characteristics, in order to isolate the independent contribution of the housing channel.

5.2.1 House prices only

Table 3: Local-area Phillips curve: house prices only

	Pre Period (2004–2020)		Post Period (2021m3–2022m6)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
$u_{m,t}$	-0.211***	-0.006	0.004	-1.388***
	(0.030)	(0.102)	(0.083)	(0.285)
$E_{m,t}[\pi_{t+1}]$	0.182***	0.115***	0.118***	-0.082**
	(0.042)	(0.039)	(0.019)	(0.040)
$p^x_{m,t}$	-0.101***	0.123***	-0.087**	0.224***
	(0.016)	(0.031)	(0.037)	(0.060)
Observations	2,163	2,079	320	320
Time FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
F-stat ($Z^u_{m,t}$)	—	86.10	—	55.57

Notes. OLS (columns 1 and 3) and IV (columns 2 and 4) estimates of equation (4). The dependent variable is the 12-month MSA-level CPI inflation rate $\pi_{m,t}$. Inflation expectations $E_{m,t}[\pi_{t+1}]$ are constructed using house prices only. $u_{m,t}$ is the negative log employment rate. $p^x_{m,t}$ is the log ratio of the manufacturing PPI to the CPI all items. In the IV specification, $u_{m,t}$ is instrumented by the Bartik shift-share $Z^u_{m,t}$; $p^x_{m,t}$ is instrumented by its 24-month lag. The OLS specification absorbs MSA and year-quarter fixed effects; the IV specification absorbs MSA fixed effects and year-quarter × initial industry-share time effects. Standard errors clustered at the MSA × year-quarter level. The pre period covers 2004m1–2020m2; the post period covers 2021m3–2022m6. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3 reports the house-prices-only specification. The qualitative pattern from Table 2 is preserved across both regimes. In the pre-COVID sample, OLS produces a steep slope ($\hat{\kappa} = -0.211$, $p < 0.01$), a positive and significant expectations coefficient ($\hat{\psi} = 0.182$, $p < 0.01$), and a negative intermediate-input coefficient ($\hat{\delta} = -0.101$, $p < 0.01$). Instrumenting unemployment and the intermediate-input price collapses the slope to $\hat{\kappa} = -0.006$

(insignificant), recovers the positive structural sign on the intermediate-input coefficient ($\hat{\delta} = 0.123$, $p < 0.01$), and leaves the expectations coefficient positive and significant at a slightly attenuated $\hat{\psi} = 0.115$ ($p < 0.01$). The first-stage F-statistic on the unemployment instrument is 86.10.

In the post-COVID sample, the OLS specification produces a flat slope ($\hat{\kappa} = 0.004$, insignificant), a positive and significant expectations coefficient ($\hat{\psi} = 0.118$, $p < 0.01$), and a negative intermediate-input coefficient ($\hat{\delta} = -0.087$, $p < 0.05$). The IV specification produces a very steep slope ($\hat{\kappa} = -1.388$, $p < 0.01$), a small *negative* expectations coefficient ($\hat{\psi} = -0.082$, $p < 0.05$), and a positive intermediate-input coefficient ($\hat{\delta} = 0.224$, $p < 0.01$). The first-stage F-statistic on the unemployment instrument is 55.57. The negative IV coefficient on the housing-only expectation measure in the post-COVID period is suggestive of a particular form of contamination: under the housing-only first-pass, the synthetic expectation series is essentially a smoothed version of MSA-level house price inflation, which during the post-COVID period was disproportionately driven by national rather than local supply shocks; once unemployment and intermediate-input prices are instrumented, the partial correlation between the housing-driven first-pass expectation and CPI inflation flips sign. The pattern reinforces the case for instrumenting the expectations regressor in the full Bayesian hierarchical implementation.

5.2.2 House prices and demographics

Table 4 reports the combined house-prices-and-demographics specification. The pattern of OLS-IV divergence is again preserved across both regimes. In the pre-COVID sample, the OLS slope is $\hat{\kappa} = -0.207$ and the IV slope is statistically zero ($\hat{\kappa} = -0.003$); the expectations coefficient is positive and significant under both estimators ($\hat{\psi} = 0.088$ under OLS, $\hat{\psi} = 0.070$ under IV); and the intermediate-input coefficient flips sign from $\hat{\delta} = -0.098$ under OLS to $\hat{\delta} = 0.128$ under IV. In the post-COVID sample, the OLS slope is essentially zero and the IV slope is $\hat{\kappa} = -1.432$ ($p < 0.01$); the OLS expectations coefficient is large and significant ($\hat{\psi} = 0.226$, $p < 0.01$) but becomes statistically zero under IV ($\hat{\psi} = -0.051$). The intermediate-input coefficient flips sign in the same direction. First-stage F-statistics on the unemployment instrument are 80.47 and 53.90 in the pre- and post-COVID samples respectively.

Table 4: Local-area Phillips curve: house prices and demographics

	Pre Period (2004–2020)		Post Period (2021m3–2022m6)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
$u_{m,t}$	-0.207***	-0.003	0.009	-1.432***
	(0.030)	(0.104)	(0.075)	(0.294)
$E_{m,t}[\pi_{t+1}]$	0.088***	0.070***	0.226***	-0.051
	(0.029)	(0.025)	(0.026)	(0.060)
$p^x_{m,t}$	-0.098***	0.128***	-0.157***	0.221***
	(0.016)	(0.031)	(0.037)	(0.073)
Observations	2,163	2,079	320	320
Time FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
F-stat ($Z^u_{m,t}$)	—	80.47	—	53.90

Notes. OLS (columns 1 and 3) and IV (columns 2 and 4) estimates of equation (4). The dependent variable is the 12-month MSA-level CPI inflation rate $\pi_{m,t}$. Inflation expectations $E_{m,t}[\pi_{t+1}]$ combine house prices and demographics. $u_{m,t}$ is the negative log employment rate. $p^x_{m,t}$ is the log ratio of the manufacturing PPI to the CPI all items. In the IV specification, $u_{m,t}$ is instrumented by the Bartik shift-share $Z^u_{m,t}$; $p^x_{m,t}$ is instrumented by its 24-month lag. The OLS specification absorbs MSA and year-quarter fixed effects; the IV specification absorbs MSA fixed effects and year-quarter × initial industry-share time effects. Standard errors clustered at the MSA × year-quarter level. The pre period covers 2004m1–2020m2; the post period covers 2021m3–2022m6. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3 Discussion

The results across Tables 2, 3, and 4 deliver a single coherent message that the choice of estimator is critical and that the bias direction reverses across the pre- and post-COVID regimes.

Pre-COVID period. OLS produces a regional Phillips curve that looks textbook—a steep negative slope, a significant expectations channel, and a coefficient on the intermediate-input price that is negative. Each piece of that picture is misleading. The negative OLS coefficient on the intermediate-input price contradicts the structural prediction that δ is the (positive) intermediate-input share, indicating that OLS is contaminated by reverse causation from contemporaneous inflation shocks to upstream input prices. Once we instrument the intermediate-input price with its 24-month lag and unemployment with the Bartik shift-share, the intermediate-input coefficient becomes positive and significant, consistent with the structural restriction. The unemployment slope simultaneously collapses to zero, recovering the flat regional Phillips curve documented by Hazell et al. (2022) in their state-level specification. The expectations channel survives the IV correction with a positive and significant coefficient, indicating that even after instrumenting unemployment and the intermediate-input price—and even without instrumenting expectations themselves—regional household inflation expectations have a quantitatively meaningful association with realized regional inflation.

Post-COVID period. The OLS picture also looks intuitive but is again misleading: a flat slope, a strong expectations channel, and a negative coefficient on the intermediate-input price. The IV-corrected picture is the opposite. The slope is very steep ($\hat{\kappa}$ between -1.34 and -1.43 across the three measures, all significant at the 1% level), the intermediate-input coefficient is positive and significant, and the OLS expectations coefficient collapses toward zero. The collapse of the expectations coefficient under IV in the post-COVID period must be interpreted with caution because expectations are not themselves instrumented in the preliminary results: when the IV strategy recovers a credible coefficient on unemployment that absorbs more of the inflation variation, the residual variation that the regression attributes to expectations under OLS is mechanically reduced. Whether the post-COVID expectations channel survives instrumentation of $E_{m,t}[\pi_{t+1}]$ is the central empirical question that the full Bayesian hierarchical IV strategy is designed to answer.

The case for instrumenting expectations. Across both regimes, the intermediate-input coefficient flips sign between OLS and IV; across both regimes, the IV strategy recovers a coefficient on $u_{m,t}$ that is qualitatively different from the OLS estimate; and the resulting estimate of $\hat{\psi}$ shifts in ways that confound the causal effect of expectations with the bias in the other coefficients. The natural and necessary next step is to instrument $E_{m,t}[\pi_{t+1}]$ with the rolling-decennial shift-share $Z^{\pi}_{m,t}$ in equation (10), paired with the Bayesian hierarchical first stage of Section 4.2 that produces $\hat{\theta}_{m,t}$ as a statistically independent regressor. The full implementation will deliver causal estimates of all three coefficients in equation (4) simultaneously and resolve the open question of whether the post-COVID expectations channel is a structural feature of the inflation surge or an artifact of OLS bias on the unemployment and intermediate-input coefficients.

6 Conclusion

The dominant approach to estimating the regional Phillips curve achieves clean identification by absorbing aggregate shocks and common long-run inflation expectations through two-way fixed effects (Hazell et al., 2022). The word *common* in that identifying assumption is doing enormous work: a parallel literature documents that inflation expectations vary systematically across U.S. metropolitan areas in ways that may be correlated with local labor market conditions, raising the prospect of substantial omitted variable bias in the leading slope estimate.

This paper has set out a strategy to quantify that bias and presents preliminary evidence on the role of regional inflation expectations across two distinct inflationary regimes. The full methodology combines a two-model Bayesian hierarchical first stage—a national-only model that produces demographic-cell time predictions used to construct an instrument, and an MSA-level multilevel model with an MSA-by-time random effect under spatio-temporal smoothing that produces the regressor—with a rolling-decennial shift-share instrumental-variables strategy that uses only the demographic-share-driven variation in expectations. The two-model structure is essential: it ensures the instrument and the regressor in the second-stage Phillips curve are statistically independent objects, breaking the circularity that would arise if both were drawn from the same posterior.

As preliminary evidence pending the implementation of the full strategy, we report OLS and instrumental-variables estimates of the regional Phillips curve on a 21-MSA monthly panel split into a pre-COVID period (2004–2020) and the post-COVID inflation surge (2021m3–2022m6), using a transparent first-pass synthetic projection of MSC microdata onto MSA-

level demographics. Across all three first-pass variants—demographic-only, house-prices-only, and house-prices-with-demographics—the comparison of OLS and IV estimates reveals that OLS misleads in opposite directions across the two regimes. Pre-COVID OLS produces a steep slope, a positive expectations channel, and a (structurally implausible) negative intermediate-input coefficient; instrumenting unemployment with a Bartik-style sectoral labor demand shock and the intermediate-input price with its 24-month lag recovers the flat slope of Hazell et al. (2022), restores the positive structural sign on the intermediate-input coefficient, and leaves the expectations channel positive and significant. Post-COVID OLS produces a flat slope, a strong expectations channel, and a negative intermediate-input coefficient; the same IV strategy produces a very steep slope ($\hat{\kappa} \approx -1.34$ to -1.43 across measures, all significant at the 1% level), a positive intermediate-input coefficient, and an expectations coefficient that becomes statistically indistinguishable from zero. The first-stage F-statistics on the unemployment instrument range from 40 to 86 across specifications.

The central message of the preliminary evidence is the quantitative importance of instrumenting both unemployment and the intermediate-input price in the regional Phillips curve. The most striking sign of OLS contamination is the sign flip on the intermediate-input coefficient, which contradicts the structural Cerrato–Gitti restriction under OLS and aligns with it under IV in both regimes. The recovery of the flat slope of Hazell et al. (2022) in the pre-COVID IV specification provides direct empirical support for the central finding of that literature, while the post-COVID IV specification uncovers a much steeper slope that is consistent with a regime-dependent regional Phillips curve and with the regime-switching evidence in Cerrato and Gitti (2022) and Gitti (2024). The disappearance of the post-COVID OLS expectations channel under IV must be interpreted with caution: because $E_{m,t}[\pi_{t+1}]$ is not itself instrumented in the preliminary results, the change in $\hat{\psi}$ between the OLS and IV columns is confounded with the bias correction on the unemployment and intermediate-input coefficients. Disentangling the true expectations channel from the bias on the other coefficients requires instrumenting expectations with the rolling-decennial shift-share described in Section 4.3.

We view the implementation of the full Bayesian hierarchical IV strategy—which delivers $\hat{\theta}_{m,t}$ as a statistically independent regressor and $Z^{\pi}_{m,t}$ as its instrument simultaneously, alongside the Bartik unemployment instrument and the lagged intermediate-input instrument used here—as the central next step. Several extensions follow naturally. First, applying the methodology to the firm side using SoFIE—and to the post-2021 household sample using ICIE directly as a non-shift-share check—would let us assess the robustness of the central findings to the particular source of regional expectations. Second, extending the

analysis to the long-run expectation horizon would let us address the original Hazell et al. (2022) identification claim about long-run anchoring directly. Third, the BIE-based unit-cost regression provides a complementary check on the price-setting margin that the structural NKPC restricts but does not separately identify. We leave these for future work.

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